



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

GENERATIVE AI AND APPLICATIONS [CSE-40A3]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Outline the necessity of Generative AI models in personalization, user engagement, and drug discovery. (5)
 - A)
 - B) Explain the limitations of Variational Autoencoders over Generative Adversarial Networks in detail. (5)
- 2) A manufacturing company aims to improve predictive maintenance by detecting anomalies in industrial machinery based on sensor data. These machines generate time-series sensor readings (e.g., temperature, vibration, and pressure) that indicate their operational status. However, since failures are rare and unpredictable, collecting labeled fault data is challenging. To address this, the team decides to use a Variational Autoencoder (VAE) to learn the normal operating patterns of the machinery and detect anomalies when deviations occur.
 - A) Develop a Python script/pseudocode using TensorFlow/Keras that: (5)
 - Load and preprocesses a dataset of time-series sensor readings from industrial machines. You can use the NASA C-MAPSS dataset (for jet engine sensor data), the CWRU Bearing Dataset (for vibration-based fault detection), or the NAB Dataset (for general time-series anomalies).
 - Builds a Variational Autoencoder (VAE) where the encoder maps input data into a latent probability distribution, and the decoder reconstructs the original input.
 - B) Use the above scenario in Question 2A and answer the following by developing a python script/pseudocode. (5)
 - Train the model using Reconstruction Loss (MSE) and Kullback-Leibler (KL) Divergence to enforce a structured latent space.
 - Evaluate the model by computing the reconstruction error and identifying anomalies when the error exceeds a predefined threshold.
- 3) A digital art company wants to develop an AI model that can generate high-quality paintings resembling famous artistic styles, such as Impressionism, Cubism, or Surrealism. Traditional GANs often suffer from instability and mode collapse, making it difficult to generate diverse and realistic artwork. To overcome these challenges, the team decides to use a Wasserstein GAN (WGAN), which improves training stability and generates more realistic images. (5)
 - A)

Develop a Python script/pseudocode using TensorFlow/Keras that:

- Loads and preprocesses a dataset of paintings, such as the WikiArt dataset or the Painter by Numbers dataset.
- Builds a Wasserstein GAN (WGAN) consisting of:
 - A Generator that takes random noise as input and generates artwork.
 - A Critic (instead of a Discriminator) that scores how real or fake an image is, using the Wasserstein loss function.

B) Use the above scenario in Question 3A and answer the following by developing a python script/pseudocode.

- Train the WGAN using the Wasserstein loss function and applies gradient penalty (WGAN-GP) to enforce the Lipschitz constraint.
- Evaluate the model by generating new paintings and comparing them with real artwork. (5)

4) A publishing company wants to develop an AI-based tool that can generate short stories or continue unfinished stories based on a given prompt. The goal is to train a model that learns storytelling patterns, sentence structures, and narrative flow from a large dataset of books and short stories. To achieve this, the team decides to use a Recurrent Neural Network (RNN) that can predict the next word or sentence based on previous text inputs.

A)

Develop a Python script/pseudocode using TensorFlow/Keras that:

- Loads and preprocesses a text dataset, such as the Gutenberg dataset (classic literature) or the Harry Potter Fan Fiction dataset (if fine-tuning for a specific genre).
- Builds an RNN-based language model, using:
 - An Embedding layer to convert words into dense vector representations.
 - One or more LSTM layers to learn sequential text dependencies.
 - A Dense layer with a Softmax activation to predict the next word or sentence.

(5)

B) Use the above scenario in Question 4A and answer the following by developing a python script/pseudocode.

- Train the model using Categorical Cross-Entropy loss and the Adam optimizer, ensuring it learns to generate coherent text.
- Evaluate the model by feeding it a prompt and generating a continuation of the story. (5)

5) Define prompt engineering and discuss its significance in improving LLM output quality.

(5)

A)

B) Illustrate the ethical considerations in Prompt engineering.

(5)

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